

AeroVanti SkyPoints — Deep Dive

The exhaustive record behind the board summary · dual-track (pandas + DuckDB)

AeroVanti SkyPoints — Deep Dive (index)

The exhaustive record behind the board summary. Audience: analysts / managers who will act on it. Every number is computed on **both tracks** (pandas + DuckDB over the same curated parquet) with a parity assertion; run `python data_analysis/parity/run_all.py` to reproduce.

The two business problems

#	Problem	Deep-dive file	Findings
A	Flash Redemption A/B test — should it roll out, and to whom?	deep_dive_A_ab_test_readout.md	01–06
B	Why members hoard points, and what it costs	deep_dive_B_program_health.md	07–08

Each finding also stands alone under [../findings/](#) with its own context · question · hypothesis · both-track method · evidence · limitations · recommendation.

Headline

Flash Redemption **works** — it lifts the 12-week redemption rate by **+2.7 pp pooled** (+3.2 pp re-weighted to the member base; $p \approx 2 \times 10^{-6}$) — but the effect is **concentrated in Blue and Silver** (arm×tier interaction $p = 0.0007$; nothing measurable for Gold/Platinum), and **roughly half the headline is a launch bump that decays** (durable $\approx +1.5$ pp). Guardrails: disengagement improves (−2.0 pp, $p = 0.001$); flight revenue is unharmed but the test is too small to *prove* non-inferiority; redemptions get **shallower** (−R\$19 per redeeming member). **Recommendation: roll out to Blue + Silver, hold Gold/Platinum, keep a revenue and a redemption-depth guardrail live.**

Data & method notes

- Analysis surface: data/curated/*.parquet (Kimball star, 18 tables). See [../00_START_HERE.md](#) and docs/05_data_model.md.
 - The experiment's true effect is **not** in any doc — it lives only in etl/config.py:FLASH_EFFECT; this read-out detects it from the data.
 - Randomisation used blocked (pair-matched) assignment within tier strata, so the balance check passes by construction (finding 02).
 - Post-window horizon is ~14 weeks: lapse guardrails are **90-day proxies**, not 12-month outcomes.
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Deep Dive A — Flash Redemption A/B test read-out

Business problem. SkyPoints members accumulate miles but rarely redeem: only ~32% of active members have ever redeemed, and non-redeemers lapse far more often. The loyalty team believes the **10,000-point minimum redemption threshold** is the common cause — a Blue member with 4,000 points can do nothing with them, never gets the "I got something" moment, and disengages. In program months 19–21 they ran **Flash Redemption** as a randomised, member-level A/B test: threshold **10,000** → **3,500** points, plus a low-cost reward catalog (seat selection, extra bag, lounge pass, gift cards, Wi-Fi vouchers).

The decision. Roll Flash Redemption out to the whole program, roll it out to some tiers, or don't. Plus: what durable lift to put in the business case, and which guardrails to keep live.

1. Design & data (docs/01 §7)

Element	Value
Unit / split	member / 50-50, blocked (pair-matched) within tier strata
Eligible	active (earn or flight in trailing 12m) and not lapsed at 2026-03-01
Sample	15,000 — Blue 7,000 · Silver 3,800 · Gold 2,600 · Platinum 1,600 (Gold/Platinum oversampled for the heterogeneity cut)
Window	12 weeks, 2026-03-02 → 2026-05-24; + 8 wk (revenue) / 90 d (disengagement) look-ahead
Primary metric	redemption rate = share with ≥ 1 redemption in the window

Element	Value
Guardrails	revenue / member · net point-liability cost / member · 90-day disengagement · redemption value per redeemer
Surface	fact_experiment_outcome (1 row/subject), fact_experiment_member_week (weekly), fact_experiment_assignment (frozen covariates)

Power: planning was control 11.0%, MDE +2.2 pp, 80% power → $\approx$ 3,450/arm. The realised design (7,500/arm, control 12.8%) gives **0.97 power at the +2.2 pp MDE** and an 80%-detectable effect of $\approx$ +1.6 pp.

2. Randomisation check (finding 02)

- **Sample-ratio mismatch:** every stratum is exactly 50/50 ($\chi^2 = 0.000$, $p = 1.000$).
- **Covariate balance:** SMD on all six frozen covariates (pre-period earn, flights, redemptions, balance, tenure, prior-redeemer flag) — **max |SMD| = 0.039**, all inside the 0.1 threshold.
outputs/figures/02_love_plot.png.
- Blocked randomisation (rank by a pre-period redemption-propensity score → adjacent pairs → coin flip per pair) balances prior behaviour by construction.

→ the effect can be read causally.

3. Primary metric (finding 03)

	control	treatment
Redemption rate	12.76% (957/7,500)	15.45% (1,159/7,500)

Estimate	Value	95% CI
Absolute lift (pooled)	+2.69 pp (+21.1% rel.)	[+1.58, +3.81] pp
z / two-sided p	4.74 / 2.2×10^{-6}	—
Post-stratified to population tier mix	+3.21 pp	[+2.20, +4.22] pp
Covariate-adjusted (logistic AME)	+2.78 pp	($p = 1 \times 10^{-7}$)

The pooled estimate is on the *oversampled* design, which under-weights Blue (largest effect); **post-stratified to the 68/20/9/3 member base the lift is $\approx$ +3 pp** — the number to quote for a full rollout.

4. Heterogeneity by tier (finding 06) — the decision hinges here

tier	control	treatment	lift (pp)	95% CI	significant
Blue	6.1%	9.9%	+3.77	[+2.50, +5.04]	yes (p = 2e-8)
Silver	10.8%	13.0%	+2.16	[+0.10, +4.22]	yes, marginal (p = 0.04)
Gold	19.4%	21.9%	+2.54	[−0.57, +5.65]	no (underpowered)
Platinum	35.6%	35.1%	−0.50	[−5.19, +4.19]	no

- **Arm × tier interaction:** LR $\chi^2 = 17.0$, df = 3, p = 0.0007 — the effect genuinely differs by tier.
- **Linear-in-tier-rank interaction:** −0.172, p = 0.0001 — a monotone decreasing gradient Blue → Platinum.
- Forest plot: [outputs/figures/06_forest.png](#).

Blue — the segment the intervention was designed for — gains +3.8 pp on a 6% base (+62% **relative**). Gold/Platinum show no measurable benefit and already redeem freely. "Not significant" for Gold/Platinum means *consistent with no effect*, not *proven zero* (both are underpowered), but the interaction test says the gradient is real.

5. Novelty effect (finding 05)

- Weekly (treatment – control) increment: **OLS slope −0.044 pp/week**, p = 0.017.
- Weeks 1–4: +0.42 pp/wk → weeks 9–12: +0.13 pp/wk.
- Cumulative 12-week lift +2.69 pp; **durable (late-rate-equivalent) $\approx$ +1.5 pp**.
- [outputs/figures/05_novelty.png](#).

Roughly half the headline is a launch bump that decays. **Build the business case on $\approx$ +1.5–2 pp durable**, expect a temporary launch spike.

6. Guardrails (finding 04)

Guardrail	Δ (treatment – control)	95% CI	p	Verdict
Revenue / member (window + 8 wk)	−R\$0.8 (−0.09%)	[−R\$48, +R\$46]	0.97	no harm, but NI vs −3% not established (one-sided p = 0.14)
Net liability cost / member	−R\$0.81	[−R\$1.78, +R\$0.16]	0.10	favourable direction, n.s.

Guardrail	Δ (treatment – control)	95% CI	p	Verdict
90-day disengagement	–2.01 pp	[–3.21, –0.81]	0.001	favourable
Value per redeeming member	–R\$19 (–6.8%)	[–R\$31, –R\$8]	0.001	watch — shallower redemptions; 9% of treatment redeemers use only the low-cost catalog

Revenue is not hurt (point estimate ≈ 0) but the sample is too small to *prove* non-inferiority — keep a revenue guardrail live. Disengagement improves (the clean causal echo of finding 08). Redemptions get shallower — Flash partly converts "no redemption" into "a small redemption"; watch the mix.

7. Decision

Roll out Flash Redemption to Blue and Silver. Hold Gold and Platinum at the 10,000-point threshold (no measured benefit; the low-cost catalog would only add reward cash cost).

- **Expected durable effect:** $\approx +1.5$ – 2 pp on the quarterly redemption rate for Blue/Silver ($\approx +3$ pp headline at launch, decaying).
- **Keep live during rollout:** (1) revenue per member vs a -3% floor; (2) share of redemptions that are low-cost-catalog-only.
- **Follow-ups:** a larger dedicated Gold test if a definitive answer is wanted; a longer post-rollout window to convert the 90-day disengagement signal into a 12-month lapse number; pair with a co-brand-card redemption nudge (Deep Dive B).

8. Limitations

- Post-window horizon ~ 14 weeks $\rightarrow$ lapse guardrails are **90-day proxies**.
- Gold/Platinum strata underpowered even at $n = 1,300 / 800$ per arm.
- Novelty vs noise: 12 weeks is short; the decay slope is significant but its magnitude is imprecise.
- Non-inferiority on revenue is **inconclusive**, not passed.
- All data synthetic and seeded (et1/config.py); benchmark ranges are directional.

Deep Dive B — Why members hoard points, and what it costs

Business problem. The SkyPoints relaunch business case assumed breakage would fall from ~40% toward 30% as members learned the program. It hasn't. A large, slow-moving point-liability sits on the balance sheet, and members who never redeem lapse far more often. This deep dive sizes the hoarding funnel, the liability and its estimate risk, and the redeemer→retention link that makes redemption a retention lever — the context for the Flash Redemption decision (Deep Dive A).

1. The redemption funnel (finding 07)

At 2026-08, of 109,458 active members:

stage	all	Blue	Silver	Gold	Platinum
balance ≥ Flash threshold (3,500)	68%	65%	72%	78%	83%
ever redeemed	32%	25%	37%	53%	70%
redeemed in last 12m	25%	19%	30%	46%	66%
redeemed last quarter	9%	6%	10%	18%	34%

The drop-off is not "can't afford to" (two-thirds clear the threshold) — it's "never start". **Blue is the problem:** a quarter have ever redeemed, 6% redeemed last quarter. This is exactly the stage and segment Flash Redemption targets, and finding 06 shows it moves the needle there.

2. Earn is dominated by the co-brand card (finding 07)

SkyPoints issued window-to-date, by source (M points):

source	M pts	share
BancoAV co-brand card	1,146	55%
legacy migration	397	19%
flights	301	14%
promo / partners / service	245	12%

The card issues more than half of all points — to members earning **with no flight attached**, who have the weakest reason to engage with the airline side of the program. That is the population that accumulates and lets points expire.

3. Breakage and the liability (finding 07)

- Points: issued 2,089M · redeemed 636M (**burn/earn 30.4%**) · **expired 228M = 10.9% of issued** (the first legacy-lot expiry wave; most window lots have not reached their 24-month expiry, so lifetime breakage is still trending toward ~40%).
- **Point liability at 2026-08: R\$25.7M gross · R\$15.4M breakage-adjusted (40%).**
- **Estimate risk:** the recognised liability moves **±R\$2.6M per 10 pp** of breakage assumption:

assumed breakage	recognised liability
20%	R\$20.6M
30%	R\$18.0M
40% (base)	R\$15.4M
50%	R\$12.9M

Flash Redemption reduces the liability at the margin (finding 04, G2: net cost per member –R\$0.81, favourable direction) by converting "will expire" points into redemptions — but the low-cost catalog is bought at cash cost, so the net is small, not a windfall.

Figure: outputs/figures/07_hoarding.png (reach by tier; liability trajectory).

4. Redeemers stay (finding 08) — observational, with the causal cross-check

12-month **engagement lapse** (no flight and no redemption for 12 months; plain "account lapse" is only ~4% because card auto-earn keeps accounts nominally alive):

	rate	n
prior redeemers	10.6%	14,108
never redeemed	21.1%	42,447
raw gap	–10.5 pp	
adjusted (logistic on tenure, tier, earn, flights, card-only)	–6.0 pp (p = 1e-31)	

Forward 12-month flight revenue: redeemers **R\$3,337/member** vs never-redeemers **R\$1,596** (+R\$1,741 raw).

A large association survives adjustment, but a latent engagement trait plausibly drives both redeeming and staying, so even –6 pp over-states causation. The clean causal signal is the experiment's **90-day disengagement guardrail: –2.0 pp, p = 0.001** (Deep Dive A §6). The true

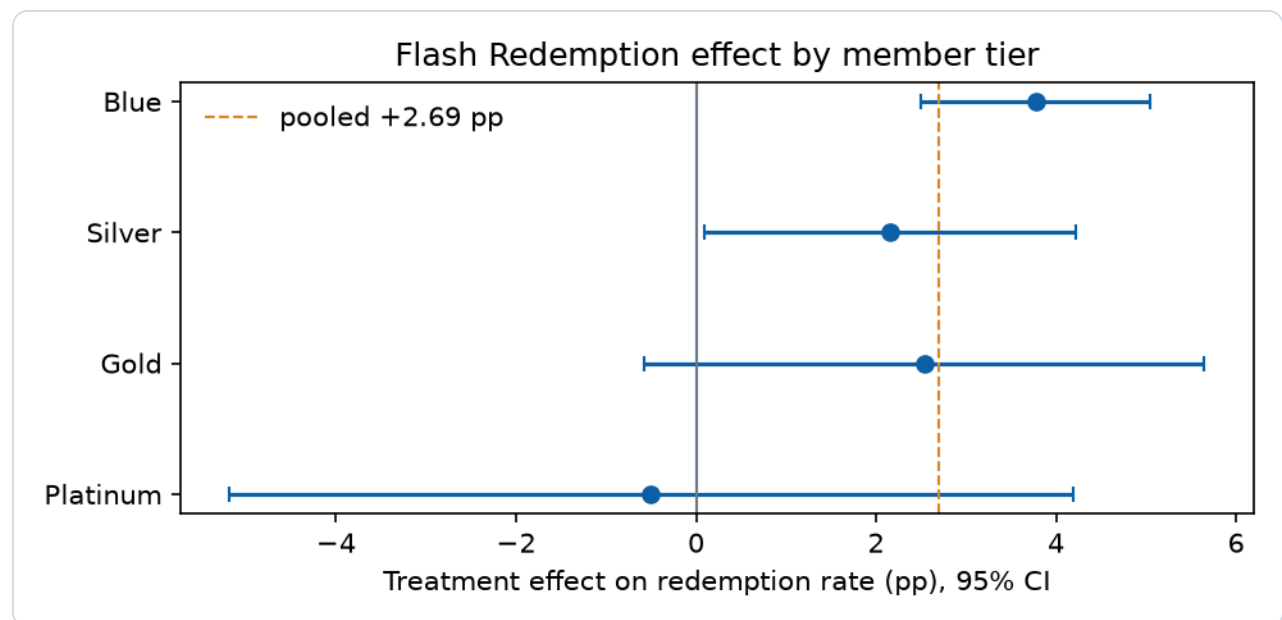
12-month retention benefit is somewhere between the experiment's short-run -2 pp and the observational -6 pp.

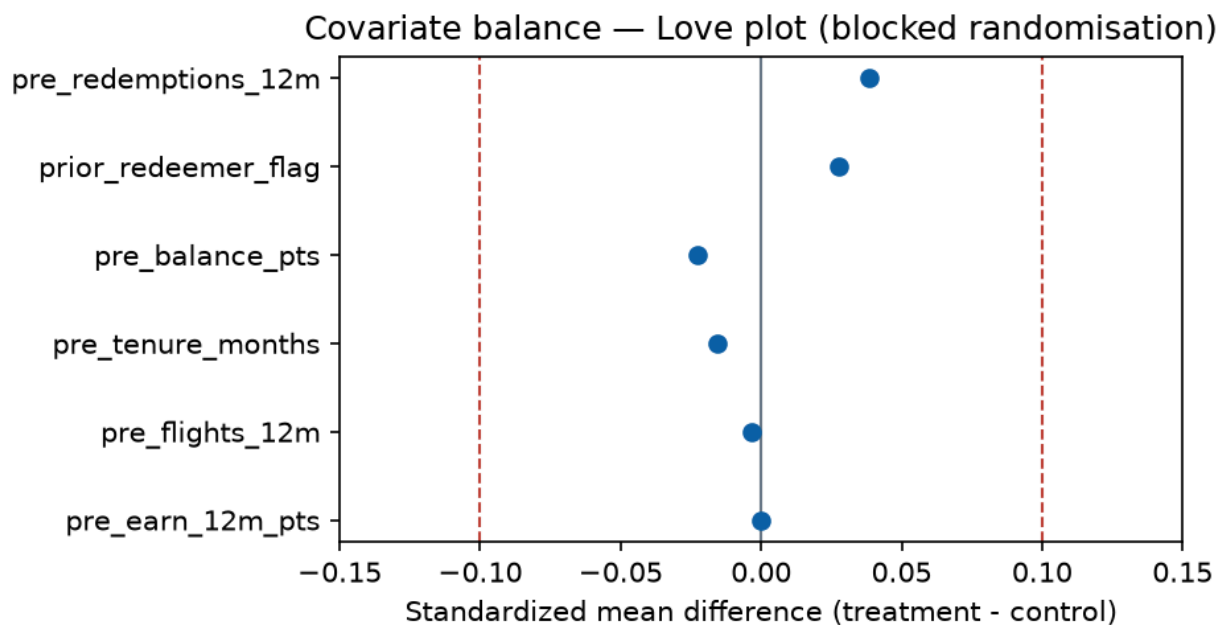
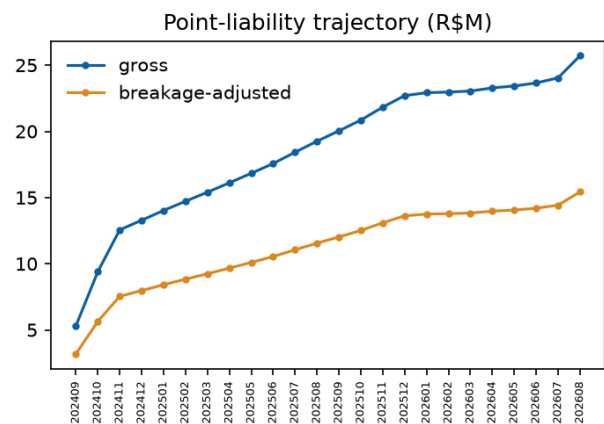
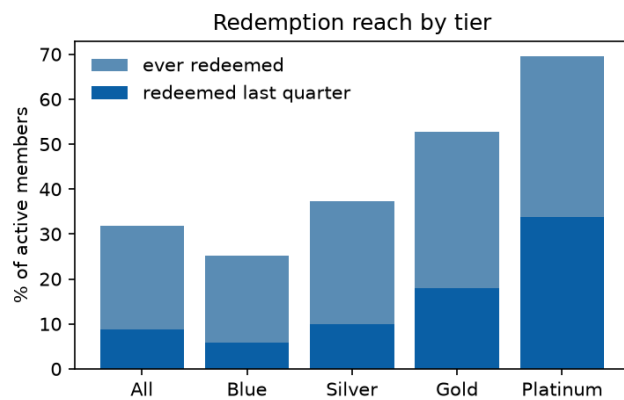
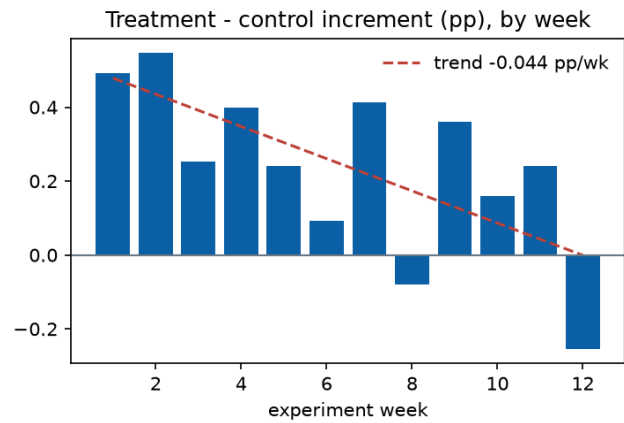
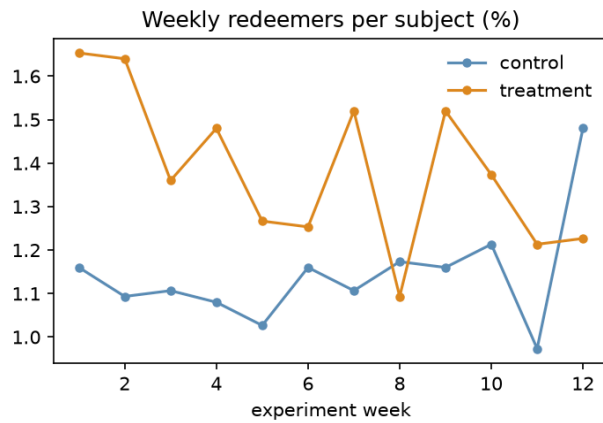
5. Limitations

- Realised breakage (11%) is window-to-date; lifetime breakage is a projection.
 - No per-lot FIFO ledger in curated (docs/08 §4) — breakage is an aggregate roll-forward.
 - Redeemer→retention is observational; confounded by latent engagement.
 - Liability unit cost (R\$0.021/pt) and the 40% breakage assumption are program inputs.
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6. Recommendations

1. **Roll out Flash Redemption to Blue + Silver** (Deep Dive A) — it attacks the funnel stage and tier where hoarding is worst.
 2. **Add a co-brand-card redemption nudge** — 55% of issuance goes to card-earn members with the weakest redemption pull; a targeted "you have enough for X" prompt is the highest- leverage adjacent move.
 3. **Run a breakage cohort study** once a full 24-month lot lifetime of data exists — the recognised-liability uncertainty ($\pm$ R\$2.6M / 10 pp) is material for the CFO.
 4. **Treat redemption enablement as a retention intervention**, sizing the benefit conservatively (closer to -2 pp than -6 pp) until a longer post-rollout window exists.
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Synthetic, seed-generated data (SEED 20260902). Every figure is computed on two tracks (pandas + DuckDB) with a parity assertion — `python data_analysis/parity/run_all.py`. The true experiment effect is not in any document; it lives only in `etl/config.py` and was detected from the data.

